

Samyuktha Krishnan

sk30@uw.edu | 425-979-6846 | www.linkedin.com/in/samyuktha-s-krishnan

EDUCATION

University of Washington Seattle, B.S in Computer Engineering, *minor in Statistics & Entrepreneurship* **Expected 2028**
Relevant Coursework: Introduction to Computer Programming, System and Software Tools, Data Structures and Parallelism, Foundations of Computing I, Foundations of Computing II, Multivariable Calculus, Matrix Algebra, Differential Equations, Statistical Computing
Activities & Societies: DubsTech (***Junior Event Coordinator***), Women in Computing (***Outreach Liaison***), Lavin Entrepreneurship Program, WE Lead, CSeed

TECHNICAL SKILLS

Programming Languages: Java, C++, Linux, Python, JavaScript, HTML/CSS, R
Frameworks & Tools: Git, React.js, Junit, Arduino, Firebase (Authentication + Firestore), PCB

WORK EXPERIENCE

Center for Industrial and Medical Ultrasound (CIMU) **Nov 2025 – Present**
Research Intern | MatLab, Git, Python, PCB

- Building and tuning ultrasonic transducers from component level (piezo stack assembly, backing layer integration, electrical interfacing)
- Developing robotic arm control logic to automate transducer placement and bonding trials
- Implementing structured experimental procedures to validate HIFU bonding performance
- Performing signal/data analysis to quantify bonding quality and acoustic output stability

Inspirit AI **Sept 2023 – Feb 2024**
Research Intern | CNNs, RNNs, LSTM, Python, Git

- Applied deep learning models (CNNs, RNNs, LSTMs) to EEG data for early seizure detection by identifying critical spatial-temporal patterns
- Built a preprocessing and augmentation pipeline, featuring band-pass filtering and signal transformations, to improve model generalization across diverse EEG inputs
- Developed a real time Python-based backend to process EEG data efficiently and enable immediate alerts for seizure prediction and intervention

PROJECTS

Custom RISC-V Core with ML Inference Engine **Aug 2025 – Jan 2026**

- Designed and simulated a low-power RISC-V core with an integrated ML accelerator, reducing inference power consumption by 25% while maintaining performance
- Implemented a verification suite that validated the core's functionality under 10 different ML workloads, ensuring robustness and reliability
- Optimized the core to execute ML models with a 15% faster execution time compared to standard RISC-V cores

Evolve Charge **Jan 2025 – Jun 2025**

- Built and tested 3 hardware prototypes of proprietary EV charging stations, cutting assembly by 20% by redesigning the internal mounting chassis
- Pitched the prototype to 20+ industry mentors and judges at STEM StartUps competition, beating out ~9 other startups to secure \$200 development grant
- Mapped out the local charging landscape to identify 5 key blind spots, resulting in a pivot toward compact installations, reducing installation costs by \$2k per site

MarineAid - Mercury Assessment and Resilience Investigation in Aquatic Domains **Aug 2023 – Apr 2024**

- Developed a Bayesian Network model for mercury resilience, achieving a 0.861 Kappa Statistic and 0.746 ROC-AUC in differentiating microhabitat recovery periods
- Implemented Shannon-Weiner Index and HRV to quantify ecosystem displacement by modeling interdependencies between climate, water chemistry, and biodiversity
- Simulated remediation node effectiveness through iterative training cycles, establishing a predictive standard for habitat resistance and long term conservation management